

WiMAX 2 OR LTE

Which Will Win The 4G Race?

If the hype is to be believed, 802.16m will redefine wireless broadband and mark the arrival of 4G. Soon after it enters the grandstands, LTE Release 10 will also be ready for the 4G race. Will one technology phase out the other or both have a place in the 4G roadmap?

■ JANANI GOPALAKRISHNAN VIKRAM

Upon being asked whether '802.16m' will be worth the hype, Prof. Ashok Jhunjhunwala shoots back, "Yes, yes. Very much. This is true wireless broadband."

Padma Shri Dr Ashok Jhunjhunwala leads the Telecommunications and Computer Networks group (TeNeT) at the Indian Institute of Technology, Madras. Through the many companies incubated by the group, he dreams of opening the world to rural India, providing them with the required information and communication technology (ICT). And 802.16m, he believes, will help advance that dream further. "Many things—education, healthcare and livelihood leveraging ICT—will come much closer to reality," he says.

802.16m: what is it?

802.16e is popularly known as WiMAX. The next version of this popular standard is on the anvil. The upcoming 802.16m air interface standard will be the basis of 'Release 2' of the existing WiMAX network standard.

802.16m has the same architectural framework and uses orthogonal frequency-division multiplexing (OFDM) techniques as 802.16e. Expected to be ratified by the Institute of Electrical and Electronics Engineers (IEEE) sometime this year, it is hyped to be true wireless broadband, and hence the beginning of 4G.



802.16m will not only mean greater capacity and other advantages for mobile service providers, it will also help provide the end users with more services and a richer mobile experience. Thanks to its superior features, it will also make it possible to provide anytime, anywhere Internet at lower costs than today.

"802.16m includes features that lead to significant improvements in cell capacity, user experience and value-added service offerings. Some of these key features are enhanced multiple-input multiple-output (MIMO) modes, better interference management, optimised channel structure, enhanced support for femtocells, introduction of relay stations, faster handovers, lower

media access control (MAC) overhead, more efficient power-saving modes, and enhanced multicast-broadcast services (MBS) and location-based services (LBS)," says Prof. Abhay Karandikar of IIT, Bombay, who participates in the IEEE 802.16m standard working group.

Other enhancements include a slightly higher spectral efficiency, lower link access latencies, emphasis on both time-division duplexing (TDD) and frequency-division duplexing (FDD) operations, multi-radio coexistence and inter-technology handover, integrated multi-hop relay, better security, self-organising and self-optimising base stations, additional target frequencies of operations and so on.